

APPENDIX 3: CLASSIFICATION SPECIFICATION

APPENDIX 3

CLASSIFICATION AND SPECIFICATIONS

Feature	Description	 easiest	 easy	 moderate	 difficult	 extreme
Tread Construction						
Minimum width of tread	The width of the trail tread	1.5m	900mm	600mm	300mm	150mm
Surface	Surface material	Stable and smooth surface. Not variable. Loose material <20mm	Mostly stable, can be variable & uneven. Loose material <30mm	Mostly stable, can be variable & uneven. Loose material <50mm	Widely variable, uneven. Loose material <100mm	Extremely variable and unpredictable. Loose material <300mm
Gradient (avg/max)	The amount of elevation change between two points over a given distance.	5% or less/8%	7% or less/15% up to 20m	10% or less/20% up to 50m	15% or less/50% up to 10m	20% or less/70% up to 20m
Line of Sight	How far ahead a rider can see the trail	Average 30m Minimum 15m	Average 20m Minimum 10m	Average 15m Minimum 7.5m	Average 15m	Average 10m
Turns	Turns are one of the most difficult areas of trail design. If turns are not designed and built exactly right they can create major issues to sustainability and lead to ongoing major maintenance requirements. Turns are control points and suitable placement of turns should be identified in Stage 4 Concept Plan. Turns should be linked into the overall design, the alignment leading into and out of a turn must be considered.					
Climbing turns	A climbing turn does not have a constructed turning platform with the apex of the turn lying directly in the fall line. Therefore these turns should be built on shallow slopes that don't exceed a grade of about 7%. Keep the turn radius as wide as possible ideally 6m. If the grade is steeper than 7% use a well-built switchback.	Min radius 4m	Min radius 3m	Min radius 2.5m	Min radius 2.5m	Min radius 2.5m
Rolling crown switchbacks	A switchback reverses the direction with the help of a level, constructed landing. Switchbacks are more difficult to build than climbing turns but are more durable on steep slopes as they do not force the rider onto the fall line. IMBA recommends the rolling crown switchback where turns occur on a near level platform that is slightly crowned to drain from all sides. Switchbacks shouldn't be built directly above one and other, they should be staggered on a hillside to prevent shortcutting and water accumulation.	Min radius 4m	Min radius 2m	Min radius 2m	Min radius 2m	Min radius 2m
Berms	Bermed or insloped turns are built up and in-sloped at a steep gradient. This helps riders maintain a smooth speed without sliding out of the turn. Side slopes with a grade less than 25% are best. The curve of the berm should be shaped for purpose and fit into the design of trail.	✗	Min radius 3m Max in slope 10% Max height 600mm	Min radius 2m Max in slope 20% Max height 1200mm	Min radius 2m Max in slope 20% Max height 2000mm	Min radius 1.5m Max in slope 40% Max height 2500mm
Vertical change						
Obstacles	Unavoidable obstacles such as logs, roots and rocks.	✗	50mm	150mm	400mm	400mm
Rollers	Rollers are a rollable feature with no sharp edges. Riders do not need to loft their bike over a roller, both wheels can be touching the ground. Rollers have different purposes: they can create a 'pumping' feeling in the trail and are good for encouraging flow, or they can be used as a slowing feature. They should be shaped fit for purpose. Each singular roller should be detailed in design as a chainage point. The spacings should be based upon approach and exit speeds, and the dimensions between rollers may change through the 'set' to suit gradients. The trail designer should space rollers sensibly considering all the conditions.	✗	Maximum height to length ratio 1:8 Max height 400mm Max 2 together Minimum length to spacing ratio 1:2 crown to crown	Maximum height to length ratio 1:7 Max height 600mm Max 3 together Minimum length to spacing ratio 1:1.5 crown to crown	Maximum height to length ratio 1:5 Max height 800mm Max of 4 together Minimum length to spacing ratio 1:1 crown to crown	Maximum height to length ratio 1:3 Max height 1000mm Max of 4 together Minimum length to spacing ratio 1:1 crown to crown
Log rollers	A log that has a ramp on both sides. The ramp may be constructed out of logs, dirt or rock. The ramp must be rollable.	✗	✗	Same as rollers	Same as rollers	Same as rollers